How robust are transcriptomic rejuvenation rankings?

Aging cohorts

Fleischer
dermal fibroblasts

GTEx
cultured fibroblasts

GTEx
skin

age effects per gene

CRISPRi Perturb-seq knockdowns

K562 RPE1 HepG2 Jurkat

18,746 knockdown profiles, 5 screens

Rejuvenation score

R = −ρ(knockdown shift, age effect)

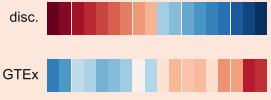
R > 0: moves expression opposite to aging

raw | proliferation-adjusted | stratified

pre-specified H1–H5 + review-round sensitivity

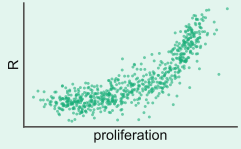
Findings

Age signature does not transfer



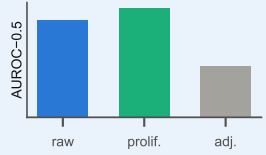
cross-cohort ρ = −0.29 to 0.26
vs split-half ρ ≈ 0.76–0.84; cohort swap
reshuffles the ranking (ρ ≈ −0.05)

Raw score ≈ proliferation



ρ(R, proliferation shift) = 0.49–0.90
only 31–48% of top candidates
survive adjustment

Evidence explained by proliferation



CellAge AUROC: raw 0.57
proliferation alone 0.58
adjusted 0.54 (near chance)

Network adds a little



STRING neighbours: ρ = 0.07–0.33
vs ~0 for degree-preserving nulls;
−0.3 to −3.4% error vs transfer